

SANDIPAN GHOSH

Senior Software Architect -- Device Routing, Placement & Agentic AI for Silicon
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SUMMARY

21-year EDA veteran who architects the **device auto-placement and routing engines** Apple, NVIDIA, and Intel depend on -- now extending them with **Agentic AI** for autonomous design closure. 7+ years as lead architect of Cadence's device routing and fill infrastructure across every major foundry node from 16nm to N2. Central designer of a patented **DRC-Aware Routing Grids System** that decouples routing algorithms from PDK-specific rules, cutting new-node enablement from months to weeks. Now building agentic systems that wrap these deterministic engines for fully automated memory chip assembly routing.

DEVICE AUTO-PLACEMENT & ROUTING -- 7+ Years as Lead Architect

DRC-Aware Routing Grids System (Patented -- US 11,790,147)

Designed the core architecture of Cadence's **Grid-Based Device Router (GBR)** -- a PDK-independent routing system for Virtuoso:

- Routing Graph:** Dynamic graph representing all available routing space across all metal layers. Edges encode DRC rules (spacing, halos, EOL). Graph updates in real-time as segments and vias are created.
- Segment Service + Via Service:** Modular DRC components that compute shadows/halos from cached constraints (minArea, minSpacing, EOL, minSpanLengthSpacing). Decoupled from routing algorithm -- enables rapid PDK enablement.
- PDK Enablement Results:** TSMC N7 (3 months), N5 (2 months), N3 (1 month), GPDK (2 weeks) -- each successive node faster due to architecture scalability.
- Multi-Net Resource Allocation:** Addressed congestion from greedy net-sequential routing where early nets consume lower-metal resources, blocking later nets.

Bayesian Optimization of Routing Parameters (Internal R&D)

Led exploration of **Optuna-based hyperparameter optimization** (Python) applied to the GBR:

- Treated router as a black box -- optimized per-design routing parameters (preferred layers, via/metal costs, net ordering) using Bayesian Optimization.
- Reduced opens from 25 to 3 on representative testcases. 57% reduction in highly-resistive metal usage.
- Validated on TSMC N3, Intel 1278 -- 100 trials on 20 parallel threads, optimal parameters at trial 60-80.
- Proposed warm-start (reuse parameters across similar PDK/designs) and one-click autonomous routing.

Device Fill, Placement & Infrastructure

- Metal & Poly Filler Engines:** Architected Virtuoso's density fill engines -- antenna fixing (jumpers, diodes), layer density analysis, DFM compliance.
- SADP Grid Infrastructure:** Grid models, checkers, and tools for Self-Aligned Double Patterning at 10nm/7nm.
- Width-Spacing Patterns** (US 10,733,351): Automatic WSP generation for color-based multi-patterning track systems, handling flipped/mirrored instances.

AGENTIC AI: Autonomous Memory Routing Engine

Production-scale routing engine orchestrated by multi-agent AI -- targeting 1M+ net capacity for memory chip assembly (DRAM/SRAM/Flash) at TSMC N2 / Intel 18A.

Component	What It Does
Memory Routing Architecture (v4.8)	Block + Periphery domain specs -- bitlines, wordlines, fishbone topology (2,155 lines)
Implementation Spec (v2.0)	RouterBase hierarchy, 14-tool RoutingToolPool, EDA callbacks for Virtuoso/Innovus/OpenROAD (2,261 lines)
Pin-to-Spine Tool (P2S v1.0)	Standalone libp2s.a -- 39 function objects, 7-phase pipeline, zero-shorts hard constraint (1,650 lines)
LangGraph Orchestrator + Meta-Optimizer	AI brain + LLM-guided Gurobi/SciPy/DEAP solver pool (2,805 lines)

Key innovation: Python/LangGraph orchestrates strategy (algorithm selection, net ordering, solver switching). C++ engine executes at microsecond-per-edge speed. Built-in DRC/LVS verification at every routing step. Zero hallucination in the compute path.

CADENCE DESIGN SYSTEMS -- 21+ Years (2004 - Present)

Role	Period	Impact
Senior Software Architect	2024 - Present	Agentic AI + device routing/fill for advanced nodes
Software Architect	2020 - 2024	Led 5-engineer team. DRC-Aware Routing Grids System (US 11,790,147). 5nm/3nm/2nm.
Sr. Member Consulting Staff	2016 - 2020	Metal/poly filler engine, antenna fixing, SADP grids for 10nm/7nm
Sr. Member Consulting Staff	2013 - 2016	Virtuoso Layout Editor: FinFET, Neighborhood Awareness, Motif-to-Qt
Member Consulting Staff	2007 - 2013	Constraint GUI (org-level award), LEF/DEF/GDS translators, 2 papers

Strategic Engagements: Apple, NVIDIA, Intel, Samsung, TSMC, Qualcomm, Broadcom, AMD, Sony, MediaTek, Amazon

OTHER AGENTIC AI PROJECTS

#	Project	Stack
1	Chat Assistant	OpenAI Agents SDK, 14 MCP servers -- E*TRADE, Gmail, Outlook, Polygon
2	MCP Servers (5)	Python, OAuth2 -- E*TRADE, Gmail, Outlook, Pushover, Proxy

3	MedExplainer	LangGraph, 7 medical APIs -- drug journey narrator (239 tests)
4	College Planner	LangGraph, 6 MCP servers -- multi-agent student guidance
5	CrewAI Portfolio (3)	Crew AI -- stock picker, backtester, portfolio manager
6	Optimization Suite	Gurobi, PuLP, SciPy, DEAP, OR-Tools -- 20+ implementations

TECHNICAL PROFICIENCIES

Domain	Technologies
Device Routing	Grid-Based Router (GBR), DRC-aware routing graphs, net ordering, congestion management, via/segment services
AI/ML Agents	LangGraph, Crew AI, AutoGen 2, OpenAI Agents SDK, MCP Architecture, Optuna (Bayesian)
EDA / Silicon	Virtuoso, SKILL, OpenAccess API, DFM, finFET/gaaFET PDKs, DEF/LEF/GDSII
Optimization	Gurobi, SciPy, PuLP, OR-Tools, DEAP (GA), Optuna, Bayesian optimization
Languages	C++, Python, SKILL, SQL
Foundry Nodes	TSMC (A14/16, N2/N2P/N3/N4/N5/N7), Intel (18A/20A/1278), Samsung (SF2/3/4/8)

CREDENTIALS

Category	Detail
Education	B.Tech, Mechanical Engineering -- Indian Institute of Technology (IIT), Delhi (Class of 2002)
Patents	7 granted (US 11790147, 10769346, 10733351, 9305133, 8612923, 8327315, 8271909) -- routing graphs, object rearrangement, multi-patterning WSPs, constraint hierarchies, neighborhood awareness
Awards	2x Organization-Level Awards, Excellence in Innovation, Excellence in Execution, Cadence Encore Award
Publications	2 papers presented at Cadence Technical Conference (US)